Applied Statistics

Lecture 7:
Jackknife and Bootstrap

Prof. Sami S. Brandt

Department of Computer Science
IT University of Copenhagen

7th October 2025

IT UNIVERSITY OF COPENHAGEN

Learning Objectives for this lecture

- Explain the Jackknife principle
- Formulate the bootstrap problem
- Apply and implement parameteric bootstrapping
- Explain the influence function and its relation to robust estimation
- Apply robust estimation in regression

Introduction

- Central element in frequentist inference is **standard error** (why?)
- Before computers, Taylor series approximations were the way of approximating the standard error
- First computer-based alternatives
  - **Jackknife** (1957), and
  - **Bootstrap** (1979).
- Accuracy estimation in robust statistics

Jackknife

- Given a vector of n observations $x = (x_1, ..., x_n)'$ from an unknown probability distribution F on a space $\mathcal{X}$.
- Define a **real-valued** statistic $\hat{\theta} = s(x)$. The statistic is just a function of x, typically another statistical estimate.

Exercise

If you are only given one sample, how would you assess the uncertainty of $\hat{\theta}$?

Jackknife

- Given a vector of n observations $x = (x_1, ..., x_n)'$ from an unknown probability distribution F on a space $\mathcal{X}$.
- Define a **real-valued** statistic $\hat{\theta} = s(x)$. The statistic is just a function of x, typically another statistical estimate.

Exercise

If you are only given one sample, how would you assess the uncertainty of $\hat{\theta}$?

- Using the prior-computer-time statistical methods, the error on $\hat{\theta}$ would require knowledge of the underlying distribution F.
- Jackknife allows you to evaluate the standard error of $\hat{\theta}$.

The Jackknife Estimate of Standard Error

The core idea is to consider how much a single element in the sample affects the estimate of $\hat{\theta}$ (cf. Leave-one-out cross-validation)

Definition

Define the $x_{(i)}$ as the sample x with the element x_i removed, or

$$x_{(i)} = (x_1, ..., x_{i-1}, x_{i+1}, ..., x_n)'.$$

Define the correspondign statistic $\hat{\theta}_{(i)} = s(x_{(i)})$

Then the **jackknife estimate of standard error** for $\hat{\theta}$ is

$$\widehat{\text{se}}_{\text{jack}} = \left[\frac{n-1}{n} \sum_{1}^{n} (\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2 \right]^{1/2},$$

where $\hat{\theta}_{(\cdot)} = \sum_1^n \hat{\theta}_{(i)}/n$.

Jackknife Estimate for the Mean

Exercise

Show that the Jackknife estimate for the mean coincides with the sample mean

$$\hat{\theta} = s(x) \equiv \bar{x}. \tag{1}$$

Features of Jackknife

- **Non-parametric**, no special form assumed for the underlying distribution F,
- Completely **automatic**,
- Hidden assumption of smooth behaviour accross sample sizes,
- **Upwardly biassed** estimate of the standard error,
- Closely connected to the Taylor series method with the difference having numerically computed directional derivatives.

Jackknife Example

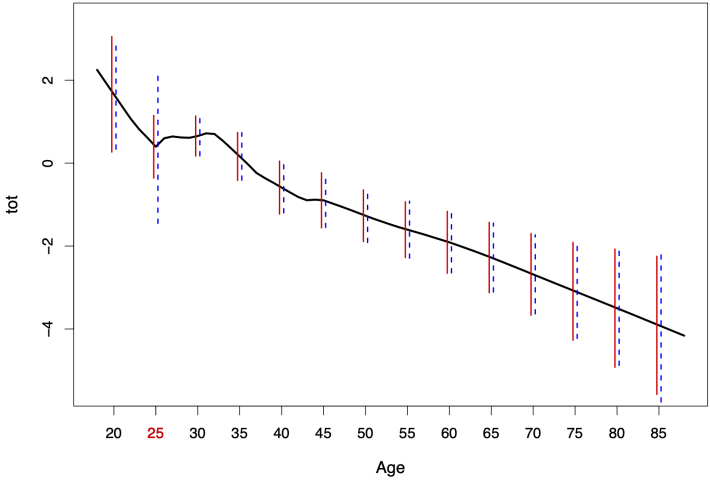


Figure 10.1 The `lowess` curve for the kidney data of Figure 1.2. Vertical bars indicate ±2 standard errors: *jackknife* (10.6) blue dashed; *bootstrap* (10.16) red solid. The jackknife greatly overestimates variability at age 25.

The bootstrap

Exercise

How can we characterise the distribution of an estimator, say the mean?

The bootstrap

Exercise

How can we characterise the distribution of an estimator, say the mean?

- Repeat the sampling several times, getting each time a different realization of the sample
- Might be unfeasible, or too expensive

The bootstrap

Idea

- Use simulation!

The bootstrap principle

- The dataset $x_1, \ldots, x_n$ is a realization of the random sample $X_1, \ldots, X_n$ from a distribution F
- Use the dataset $x_1, \ldots, x_n$ to define a distribution $\hat{F}$
- $x_1^*, \ldots, x_n^*$ is a realization of the random sample $X_1^*, \ldots, X_n^*$ from $\hat{F}$
- Approximate the distribution of the estimator $h(X_1, \ldots, X_n)$ by that of $h(X_1^*, \ldots, X_n^*)$

The bootstrap

There are two types of bootstrap, depending on how you define the distribution $\hat{F}$ from the data you have available

- Empirical (non-parametric) bootstrap
- Parametric bootstrap

Empirical bootstrap

- You have no knowledge about the distribution from which the random sample is from

Empirical bootstrap

- You have no knowledge about the distribution from which the random sample is from
- We know that the Empirical Distribution Function is a good approximation for the *true* distribution function.

- Use the Empirical Distribution Function to define $\hat{F}$
- In practice this means that you sample uniformly at random, *with replacement*, from the dataset.

Empirical bootstrap

The original estimate is obtained according to the diagram

$$F \xrightarrow{\text{i.i.d}} x \xrightarrow{s} \hat{\theta} \tag{2}$$

The bootstrapped replications are obtained in an analogous process characterised by

$$\hat{F} \xrightarrow{\text{i.i.d}} x^* \xrightarrow{s} \hat{\theta}^* \tag{3}$$

Empirical bootstrap for the mean

- We consider the distribution of $\bar{X} - \mu$
- Compute $\bar{x}$, the sample mean of $x_1, \ldots, x_n$
  1. Sample with replacement $x_1^*, \ldots, x_n^*$ from $x_1, \ldots, x_n$ uniformly at random
  2. Compute the centered sample mean $\bar{x}^* - \bar{x}$ **(*)**

  Repeat these two steps many times
- By the bootstrap principle, the distribution of $\bar{X}^* - \bar{X}$ **(*)** resembles the one of $\bar{X} - \mu$
- Can be used for several statistics, like
  - Centered sample median $Med(X_1, \ldots, X_n) - F^{\text{inv}}(0.5)$
  - Normalized sample variance S_n^2/σ^2

(*) note that in the course book here one uses the mean of the bootstrapped samples instead!

Application of the Empirical Bootstrap

- We can now evaluate quatities such as $P(|\bar{X}_n - \mu| > \epsilon)$ by approximating it by $P(|\bar{X}_n^* - \bar{x}_n| > \epsilon)$
- For instance, if the we can *ask* what is the probability of the estimator being more than $\epsilon = 5$ units away from the mean.

Properties of Bootstrap in Standard Error Estimation

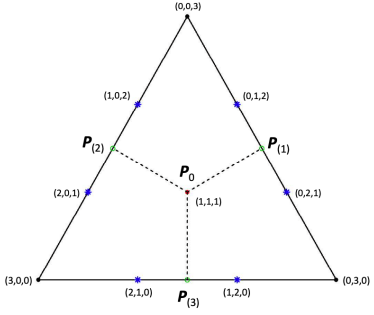
- Automatic method
- One-sample nonparametric bootstrap
- Produces non-local deviations of x^*
- Usually around 200 repetitions are sufficient for evaluating standard errors. 1000-2000 for confidence intervals.

Resampling plans

Resampling vector

- $P = (P_1, P_2, ..., P_n)'$, with $P_i \geq 0$ and $\sum_{i=1}^{n} P_i = 1$.
- Bootstrap sampling corresponds to $P_0 = (1, ..., 1)/n$.
- Jackknife corresponds to $P_{(i)} = (1, .., 1, 0, 1, ..., 1)'/(n-1)$, with the zero at the ith position.

Example: $n = 3$



Resampling plans (2)

Many other resampling schemes have been proposed, e.g.,

- The Infinitesimal Jackknife
- Multisample Bootstrap
- Moving Blocks Bootstrap
- The Bayesian Bootstrap

Parametric bootstrap

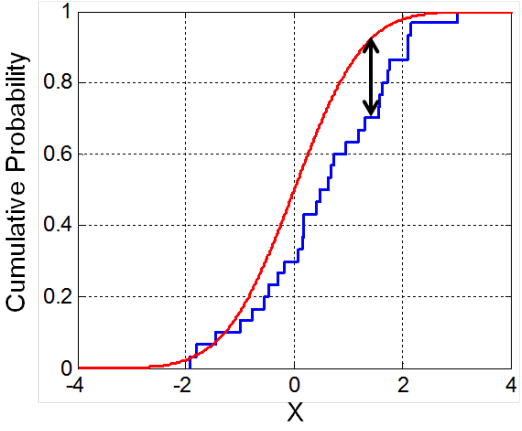
We are using a statistical model, i.e., we are using a distribution with parameters

- We can use the estimates μ of the parameters to get an estimated distribution $F_{\hat{\mu}}$
- Sample $x_1^*, \ldots, x_n^*$ from $F_{\hat{\mu}}$
- Evaluate the desired statistic $\hat{\theta}^*$ using the sample $x_1^*, \ldots, x_n^*$

Parametric bootstrap: does our model fit the data?

Example

- Kolmogorov-Smirnov distance
- $t_{ks} = \sup_{a \in \mathbb{R}} |F_n(a) - F_{\hat{\mu}}(a)|$
- t_{ks} is itself a random variable
- how likely is it to see t_{ks} if the random sample were truly distributed as we hypothesize?



Parametric bootstrap: does our model fit the data? (2)

Example

The data $x_1, \ldots, x_n$ are realizations from a distribution F_μ, where μ represets the distribution parameters.

1. Estimate the parameters $\hat{\mu}$ of the distribution using the data.
2. Draw the sample $x_1^*, \ldots, x_n^*$ from $F_{\hat{\mu}}$, estimate the parameters $\hat{\mu}^*$ using the drawn sample.
3. Compute the KS distance between the samples and $F_{\hat{\mu}^*}$
4. Repeat from step 2 many times
5. Compare the obtained KS distances to the KS distance between the original data and $F_{\hat{\mu}}$.

Book's example on Parametric Bootstrap

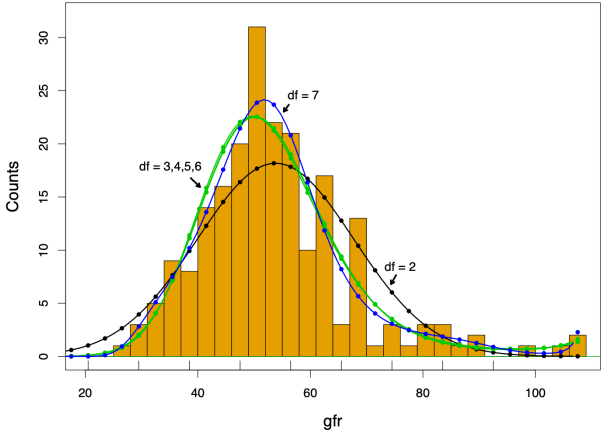


Figure 10.5 The `gfr` data of Figure 5.7 (histogram). Curves show the MLE fits from polynomial Poisson models, for degrees of freedom df $= 2, 3, \ldots, 7$. The points on the curves show the fits computed at the centers $x_{(j)}$ of the bins, with the responses being the counts in the bins. The dashes at the base of the plot show the nine `gfr` values appearing in Table 10.2.

Book's example on Parametric Bootstrap (2)

Table 10.2 *Bootstrap estimates of standard error for the* `gfr` *density. Poisson regression models* (10.54)–(10.55), df $= 2, 3, \ldots, 7$, *as in Figure 10.5; each* $B = 200$ *bootstrap replications; nonparametric standard errors based on binomial bin counts.*

	Degrees of freedom						Nonparametric standard error
`gfr`	2	3	4	5	6	7	
20.5	.28	.07	.13	.13	.12	.05	.00
29.5	.65	.57	.57	.66	.74	1.11	1.72
38.5	1.05	1.39	1.33	1.52	1.72	1.73	2.77
47.5	1.47	1.91	2.12	1.93	2.15	2.39	4.25
56.5	1.57	1.60	1.79	1.93	1.87	2.28	4.35
65.5	1.15	1.10	1.07	1.31	1.34	1.27	1.72
74.5	.76	.61	.62	.68	.81	.71	1.72
83.5	.40	.30	.40	.38	.49	.68	1.72
92.5	.13	.20	.29	.29	.34	.46	.00

Introduction to Influence Function

- The **Influence Function** measures the sensitivity of an estimator to small changes in the data.
- It assesses how much a single observation affects the estimator.
- Useful in **robust statistics** to understand how estimators respond to outliers or extreme values.

Definition of Influence Function

Formal Definition

The influence function of an estimator T at a point x with respect to a distribution F is:

$$IF(x; T, F) = \lim_{\epsilon \to 0} \frac{T((1-\epsilon)F + \epsilon\delta_x) - T(F)}{\epsilon}$$

Where:

- $T(F)$ is the estimator applied to the distribution F.
- δ_x represents a point mass at x (adding x to the data).

Interpretation of Influence Function

- The influence function tells how much the estimator T changes when a small contamination is introduced.
- **Large Influence**: High sensitivity to observations at x (non-robust).
- **Small/Bounded Influence**: Less sensitivity to extreme values, meaning the estimator is more robust.

Examples of Influence Functions

Example

Mean

$$IF(x; \hat{\mu}, F) = x - \mu$$

- The influence function of the mean is linear, hence **non-robust** (outliers have a large effect).

Example

Median

$$IF(x; \hat{\mu}_{\text{med}}, F) = \frac{1}{2f(\mu_{\text{med}})} \cdot \text{sign}(x - \mu_{\text{med}})$$

- The median has a bounded influence function, making it **robust** to outliers.

Importance in Robust Estimation

- The influence function helps in designing estimators that are less sensitive to outliers.
- **Classical estimators** (mean, variance) have unbounded influence functions.
- **Robust estimators** (Huber's M-estimator, trimmed mean) are designed with bounded influence functions.

Book's example on Influence Functions

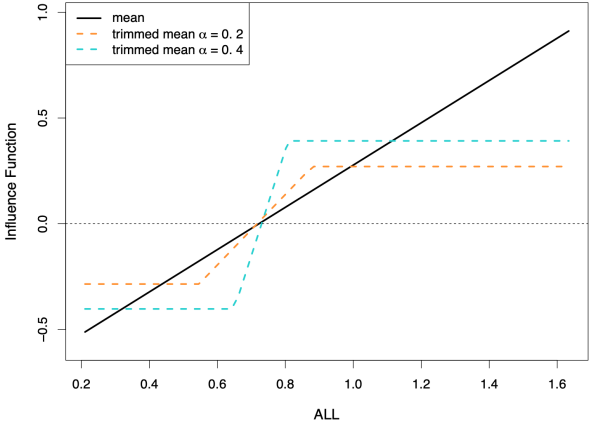


Figure 10.7 Empirical influence functions for the 47 leukemia `ALL` scores of Figure 1.4. The two dashed curves are $\mathrm{IF}_\alpha(x)$ for the trimmed means (10.68), for $\alpha = 0.2$ and $\alpha = 0.4$. The solid curve is $\mathrm{IF}(x)$ for the sample mean $\bar{x}$ (10.60).

Book's example on Influence Functions (2)

Table 10.3 *Trimmed means and their bootstrap standard deviations for the 47 leukemia* `ALL` *scores of Figure 1.4;* $B = 1000$ *bootstrap replications for each trim value. The last column gives empirical influence function estimates of the standard error, which are also the infinitesimal jackknife estimates* (10.41). *These fail for the median.*

	Trim	Trimmed mean	Bootstrap sd	(IFse)
Mean	.0	.752	.040	(.040)
	.1	.729	.038	(.034)
	.2	.720	.035	(.034)
	.3	.725	.044	(.044)
	.4	.734	.047	(.054)
Median	.5	.733	.053	

Applications of Influence Functions

- Used in **Outlier Detection** to assess the impact of individual points.
- Helps in evaluating the **robustness** of models.
- Applied in **regression models** to assess the impact of influential data points (e.g., leverage in linear regression).